



Crash-based quantitative trading strategies: Perspective of behavioral finance

Yan Fang^a, Jie Yuan^a, J. Jimmy Yang^{b,*}, Shangjun Ying^{a,1}

^a Shanghai University of International Business and Economics

^b Oregon State University, School of Accounting, Finance, and Information Systems, College of Business, 426 Austin Hall, Corvallis, OR, 97331 United States

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ABSTRACT

Inspired by the studies on stock market crashes, we use documented indicators from behavioral finance to construct two quantitative trading strategies, i.e., Crash + Timing Strategy and Crash + Momentum-Reversal Strategy. Empirical analyses show that both strategies are effective and robust. Behavioral factors can be beneficial to investors when they are incorporated into their trading strategies.

1. Introduction

With the development of the behavioral finance, many scholars seek to explain the causes of stock market crashes using behavioral theories. For instance, Avery and Zemsky (1998) point out that herding can lead to mispricing of assets and price bubbles. Kim et al. (2016) suggest that firms with overconfident CEOs have higher stock price crash risk than other firms. Jang and Kang (2019) find that price crashes could be caused by the overpricing driven by institutional investors. While existing studies focus on exploring the influential factors that may lead to stock market crashes or predicting crash events in stock markets, we investigate how investors can profit from those events.

Based on the aforementioned studies, a price crash can be caused by investors' cognitive biases, such as herding bias and overconfidence bias. Even though such cognitive biases can result in mispricing of securities, investors may theoretically expect excess returns generated from a stock market crash as long as they rationally use certain behavioral deviations (Conrad et al., 2014; Jang and Kang, 2019; Pelster, 2020). Therefore, it is highly possible to realize excess returns from a stock market crash.

During a stock market crash, stock prices typically move in a similar pattern. First, stock prices continue to rise for an extended period until the bubble reaches its peak. Then, the bubble bursts and the market collapses with a sudden slump. If one wants to profit from a stock market crash, the ability to select stocks and set the timing is critical. Accordingly, this paper adopts the behavioral finance approach to propose two quantitative trading strategies based on a documented crash factor. One strategy is built on the crash factor and market timing (i.e., Crash + Timing Strategy, CTS), while the other is constructed by applying the crash factor and momentum-reversal strategy (i.e., Crash + Momentum-Reversal Strategy, CMRS). Our empirical analyses present stable returns and

* Corresponding author.

E-mail address: jimmy.yang@bus.oregonstate.edu (J.J. Yang).

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confirm the effectiveness and robustness of these market crash-based trading strategies.

2. Construction of trading strategies

2.1. Crash + timing strategy

According to the concept of mean reversion and the gambler's fallacy (Tversky and Kahneman, 1971), stock prices can rebound from the current crash and possibly experience another crash. Investors can generate profits by buying the dips during the crash and selling the rips during the consequent rebound. We use the crash factor² of Jang and Kang (2019) to screen stocks. Specifically, top 10% stocks at the end of the previous month ($t-1$) and the current month (t) are sorted out by the crash factor, and the top 10% stocks at the end of the current month are regarded as stocks whose bubbles just burst. However, stocks selected solely based on crash factor could again take a nosedive, so they cannot be directly put into quantitative investing.³ We consider appropriate timing for buying and selling (i.e. market timing) by constructing the crash + timing strategy.

Due to the self-serving memory bias, investors generally repeat their previous behaviors for a risky stock position (Gödker et al., 2020). Therefore, we assume that before a stock market crashes, stocks will display characteristics similar to those perceived during the last crash. The strength/weakness of buying and selling on long and short positions before and after a crash would be the best indicator for market timing measurement. Consequently, we use the relative strength index (RSI) (Wilder, 1978) to identify the best times to buy stocks, and apply a momentum factor, which is defined as the corresponding 5-day momentum at the peak from last month, to capture the sell signals. During an observation period, we sell stocks when the momentum factor is similar to the peak momentum factor in the previous month ($t-1$). The specific process of CTS is given in Algorithm 1.

2.2. Crash + momentum-reversal strategy

Momentum (Jegadeesh and Titman, 1993) and reversal are two well-known anomalies within the field of behavioral finance. However, momentum strategies suffer from occasional large crashes (i.e. momentum crash) (Barroso and Santa-Clara, 2015; Daniel and Moskowitz, 2016). During a momentum crash, momentum often tends to reverse. That is, the portfolio with the lowest momentum becomes the best performer, and the momentum-reversal strategy realizes excess returns. Therefore, the momentum-reversal strategy can considerably enhance investment efficiency.

The second strategy is to combine the crash factor with momentum-reversal strategy. We first add the crash factor to the momentum-reversal strategy, adopt a scoring method to construct a new comprehensive scoring factor, and use this new factor to screen stocks. The stocks selected are featured with crashes in current month and expected to rebound later. That is, underperformed stocks will likely outperform in the future (Bali et al., 2011). The process of CMRS is shown in Algorithm 2.

3. Empirical analyses

We perform empirical analyses using daily stock data from January 1, 2018 to December 31, 2018 and the Center for Research in Security Prices (CRSP) value-weighted index as the market index. The sample includes all stocks with share codes 10 and 11 from NYSE, AMEX and NASDAQ. All data are from the CRSP database. The data processing and calculation of crash factor are conducted based on Jang and Kang (2019) (See Appendix 1).

3.1. Performance of CTS

To determine when to buy stocks, we choose the 14-day RSI⁴ within the oversold level as the buy limit price. An RSI below 30 (Wilder, 1978) would suggest a buy. Since a rebound may happen or continue beyond the current month, we use the next two months as the observation period. In addition, we consider the following four methods to determine the sell threshold.

Method 1: Use RSI as the benchmark and the overbought level (RSI=70) as a threshold;

Method 2: Adopt the pre-set take profit level (i.e., 3%)⁵ as a threshold;

Method 3: Utilize the highest cumulative return over the last month captured by the momentum factor as a threshold;

Method 4: Consider the highest cumulative return over the last month captured by the momentum-information discreteness (Da et al., 2014) as a threshold.

When adopting different methods to identify the best times to sell stocks, the stop-loss order is executed at 2.5% below the current stock price (Han et al., 2016). Evaluating the effectiveness of a strategy is essential for an investment. We evaluate our strategies based

² The crash factor is defined as the probability in the event of log returns lower than -70% over the next 12 months.

³ We do not discuss profit taking from short selling herein due to multiple short-selling bans in the world stock markets.

⁴ In order to choose the appropriate timeframe, a comparison is conducted among the respective performance of 14-day RSI, 24-day RSI and market index, seeking the timeframe that outperforms the market index.

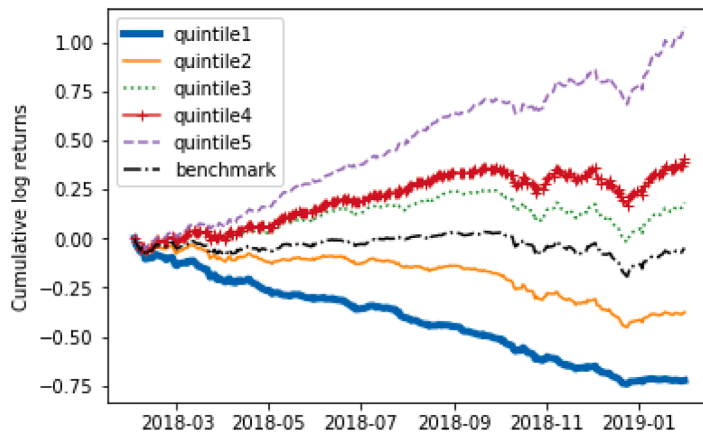
⁵ This is how to avoid emotional investing and selling into corrections within the field of behavioral finance.

Table 1

Performance of CTS with different methods in 2018.

	Method 1	Method 2	Method 3	Method 4	Market Index
Annualized return	−7.145%	3.296%	8.464%	5.515%	−10.017%
Win rate	30.113%	56.000%	37.113%	45.455%	—

Note: The sell signal in Method 1 is 14-day RSI above 70. The sell signal in Method 2 is when the cumulative return is 3% above the purchase price. The sell signals in Method 3 and Method 4 are the 5-day momentum value and the 5-day information discreteness that are equal to or above the highest cumulative return over the last month, respectively.

**Fig. 1.** Performance of CMRS from quintile 1 to 5 for 2018, and the benchmark refers to the cumulative return of the market index.**Table 2**

This table reports the annualized return, Sharp ratio, maximum drawdown and win rate on quintile portfolios sorted on the comprehensive scoring factors. At the end of each month i , quintile portfolios are constructed by sorting stocks based on the comprehensive scoring factors. Column quintile 1 is for the highest comprehensive scoring factors quintile, while column quintile 5 shows the results for the lowest comprehensive factors quintile. Column CRSP is for the performance of CRSP value-weighted index.

Panel 1: CMRS						
	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5	CRSP
Annualized return	−71.570%	−36.926%	17.761%	39.228%	104.276%	−6.012%
Sharp ratio	−0.361775	−0.151652	0.064567	0.138178	0.328911	−0.018106
Max. drawdown	−74.372%	−45.374%	−21.431%	−14.036%	−9.426%	−20.960%
Win rate	13.878%	17.809%	26.712%	31.133%	40.288%	—
Panel 2: Momentum-reversal strategy						
	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5	CRSP
Annualized return	−80.798%	−41.240%	−7.102%	46.513%	287.544%	−6.012%
Sharp ratio	−0.501626	−0.183588	−0.022715	0.156668	0.517607	−0.018106
Max. drawdown	−82.073%	−48.318%	−24.038%	−13.444%	−7.187%	−20.960%
Win rate	4.147%	9.704%	27.437%	38.024%	45.663%	—

on the annualized return and win rate, where the win rate refers to the ratio of the number of stocks with a positive return to the number of stocks with non-zero returns. The results are shown in Table 1, where the assessments for Methods 1–4 are given in columns 2–5 and the results of market index is given in column 6.

In Table 1, we can see that 1) the market return in 2018 is negative and all four Methods outperform the market index; 2) the annualized return in Method 1 is negative, and its win rate is the lowest, which implies that the market timing strategy solely depending on RSI is insufficient⁶; 3) Method 2 has the highest win rate, which indicates that the concept of take-profit can effectively avoid making wrong investment decisions, and achieves good performance in quantitative investing; 4) the annualized returns in both Methods 3 and 4 are positive and are substantially higher than the market return, which suggests that the momentum factor is a

⁶ In addition to the sell threshold of RSI=70 suggested by Wilder (1978), we also use RSI=80 as the sell threshold and RSI=20 as the buy threshold. The annualized return and win rate of Method 1 are both improved. Specifically, the annualized return improves from −7.145% to 7.465%, while the win rate increases from 30.113% to 34.408%.

Algorithm 1

The process for the CTS.

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- 1 Start with time t ,
 - a) Pick up top 10% stocks based on crash factors at the end of month t and month $t - 1$ and name them as Set 1 and Set 2, respectively;
 - b) Remove stocks which appear on both month $t - 1$ and month t from set 1, and use the remaining stocks to create a new portfolio.
 - c) For stocks listed in our new portfolio,
 - i we will buy when they reach the buy limit order during our observation period;
 - ii we will sell when they reach the sell limit order or when the stop-loss price has been reached during our observation period; otherwise, we will sell them at the end of the observation period.
 - 2 Repeat Step 1.
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Algorithm 2

The process of CMRS.

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- 1 Start with time t ,
 - a) Buy the top 20% of stocks sorted by the new comprehensive factor at the end of month t ;
 - b) Hold stocks till the end of month $t + 1$ and sell them.
 - 2 Repeat Step 1.
-

workable market timing indicator.

3.2. Performance of CMRS

In order to evaluate the performance of CMRS, all momentum factors and crash factors at the end of the month (t) are sorted in ascending and descending order, respectively, and the comprehensive scoring factors are calculated according to the scoring method.⁷ The comprehensive scoring factors are sorted in descending order, and divided into five groups (Fama and French, 1993). We buy the stocks in these five groups in month t and sell them at the end of the month ($t + 1$). Finally, we calculate the cumulative return in each group. The results are shown in Fig. 1, where each curve represents the cumulative log return of a quintile regression group.

Despite of some fluctuations in the market of 2018, the cumulative return of each group shows a clear monotonic pattern. Among them, the cumulative returns of quintiles 3 to 5 are all higher than the cumulative return of the market index, which indicates that the trading strategy brings investors an excess return. We also compare and analyze the performance of both CMRS and the traditional momentum-reversal strategy by using four indicators, i.e., annualized return, Sharpe ratio, maximum drawdown and win rate. The results are presented in Table 2.

Both strategies show strong portfolio performance. All indicators in quintiles 1–5 show clear monotonic patterns. CMRS outperforms the momentum-reversal strategy in quintiles 1–3. CMRS also outperforms the market index in quintiles 3–5. Although the best performer Quintile 5 of CMRS falls behind the control group of the momentum-reversal strategy, this trading strategy is overall an effective stock selection strategy.

Generally speaking, momentum crash is the most important factor that restricts application of the momentum strategy, and it can sometimes cause the momentum factor to fail. Although the market fluctuated in 2018, there was no momentum crashes. That is the main reason that we cannot find an obvious advantage of our proposed strategy in Table 2. To show that CMRS is effective during a momentum crash, we examine the data from January 1, 2015 to December 31, 2015 (momentum crashes happened over this period) to test the robustness of our trading strategies. The results show that both CTS and CMRS achieve favorable results, but the momentum-reversal strategy generally fails. Please see Appendix 2 for details.

In addition, we examine an up market period from January 1, 2019 to July 31, 2019 to test whether our trading strategies would work when the overall market return is positive. We find that the CTS strategy performs even better in the up market as annualized returns and win rates in 2019 are higher than those in 2018 and 2015. Results from the CMRS strategy are similar to those in 2018. Those results are not tabulated, but they are available on request.

4. Conclusion

Inspired by the latest studies on stock market crashes, this paper constructs two quantitative trading strategies and tests their performances. Empirical analyses show that 1) the crash factor cannot directly function as a momentum factor, but when the crash factor combines with an appropriate timing indicator, it could generate excess returns for investors; 2) the momentum factor is useful in a market timing approach; 3) during a volatile market, a momentum-reversal strategy may fail while the CMRS proposed in this paper is still effective.

Unlike previous studies, this article explores quantitative trading strategies from the perspective of behavioral finance, considering

⁷ The so-called scoring method is a method where stocks' crash factors are ranked in ascending order, their momentum factors are ranked in descending order, and the average of the two ranks is defined as the comprehensive scoring factor.

Table A.1

Performance of CTS with different methods in 2015.

	Method 1	Method 2	Method 3	Method 4	Market Index
Annualized return	−4.749%	9.497%	6.332%	0.712%	−6.989%
Win rate	28.049%	57.927%	46.154%	43.357%	—

the effect from stock market crash, momentum factor, and RSI. This approach provides investors useful trading strategies to potentially exploit behavioral bias in stock markets. However, there is room for future studies to further explore and extend these strategies. First, more studies are needed in the application of those strategies. For instance, potential profits from long-short portfolios in the context of the second strategy can be explored. Second, the construction of the two strategies only takes into consideration the behavior of short-term investors. A long-term approach can provide a complete picture and strengthen our understanding of market crash-based trading strategies. Third, we use the market index as the benchmark to measure performance of our trading strategies, but risk-adjusted returns can be used to appropriately account for risks involved in the trading strategies.

Declarations of Competing Interest

None

Appendix 1. Crash factor

Crash factor is defined as:

$$\text{Crash factor} = \frac{\exp(C)}{1 + \exp(C) + \exp(J)}$$

where

$$C = \alpha_0 + \alpha_1 \text{RM12} + \alpha_2 \text{EXRET12} + \alpha_3 \text{TVOL} + \alpha_4 \text{TSKEW} + \alpha_5 \text{SIZE} + \alpha_6 \text{DTURN} + \alpha_7 \text{AGE} + \alpha_8 \text{TANG} + \alpha_9 \text{SALESG}$$

and

$$J = \beta_0 + \beta_1 \text{RM12} + \beta_2 \text{EXRET12} + \beta_3 \text{TVOL} + \beta_4 \text{TSKEW} + \beta_5 \text{SIZE} + \beta_6 \text{DTURN} + \beta_7 \text{AGE} + \beta_8 \text{TANG} + \beta_9 \text{SALESG}$$

Variable RM12 is the log return of the CRSP value-weighted index over the past 12 months; EXRET12 is the log return over the past 12 months in excess of RM12; TVOL and TSKEW are the standard deviation and skewness, respectively, of daily log returns over the past 6 months; SIZE is the log of market capitalization; DTURN is the detrended turnover, defined by the six-month average of monthly share turnover minus the prior 18-month average; AGE is the number of years since the firm's first appearance on the CRSP monthly stock file; TANG is tangible assets divided by total asset; and SALESG is the sales growth over the prior year. Parameters α_i and β_i ($i = 0, 1, \dots, 9$) are from Jang and Kang (2019).

Appendix 2. Robustness testing for quantitative trading strategies

What does the advantage of the proposed trading strategies lie in, chance or strength? We further test their reliability and robustness. Data from CRSP database over the period from January 1, 2015 to December 31, 2015 are used for the robustness testing.

1. Robustness Test for CTS

To identify the best times for selling stocks, the 24-day RSI within the overbought level is used as the buy limit price, and the stop-loss order is set to be 2.5% below the current stock price. The strategy's performances for four methods defined in Section 3 are given in Table A.1.

As we can see, 1) over the period, the market return is negative and all the Methods 1–4 outperform the market index; 2) the annualized return in Method 1 is negative and its win rate is the lowest, indicating that the market timing strategy solely depending on RSI is insufficient; 3) Method 2 has the highest annualized return and win rate, implying the take-profit strategy can effectively avoid

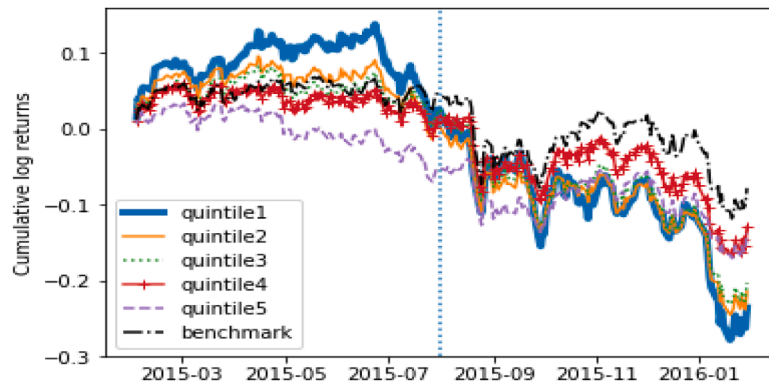


Fig. A.1. Performance of CMRS from quintile 1 to 5 for 2015 CMRS. Note: Benchmark refers to the cumulative return of the market index. The vertical line represents the date of July 31, 2015.

Table A.2

Performance of CMRS and Momentum-reversal Strategy in 2015.

Panel 1: CMRS						
	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5	RM
Annualized return	−23.039%	−20.981%	−19.836%	−12.608%	−13.447%	−7.620%
Sharp ratio	−0.073	−0.082	−0.078	−0.052	−0.062	−0.027
Max. drawdown	−36.575%	−31.096%	−29.028%	−21.128%	−19.734%	−17.486%
Win rate	45.117%	22.472%	22.532%	23.088%	22.298%	—
Panel 2: Momentum-reversal strategy						
	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5	RM
Annualized return	−20.144%	−17.747%	−15.981%	−15.022%	−21.312%	−7.620%
Sharp ratio	−0.064	−0.070	−0.065	−0.064	−0.089	−0.027
Max. drawdown	−34.123%	−27.822%	−24.707%	−22.425%	−27.636%	−17.486%
Win rate	44.826%	22.170%	23.241%	22.951%	22.182%	—

making wrong investment decisions; 4) the results in both Methods 3 and 4 considerably outperform the market index, which indicates that the momentum factor is a workable market timing indicator. In sum, this trading strategy seems robust.

2. Robustness Test for CMRS

Data in 2015 are divided into groups, since there exists a crash. The cumulative return of each group is shown in Fig. A.1.

In a relatively stable market (before August 2015), the cumulative return of each group has a clear monotonic pattern. The cumulative returns of quintiles 1 to 3 are higher than the cumulative return of the market index.⁸ When the market experiences fluctuation (since August 2015), the cumulative return of each group shows no clear monotonic pattern. The performances of both CMRS and the momentum-reversal strategy are shown in Table A.2.

Although all groups underperform the market index, the annualized returns and maximum drawdowns of CMRS maintain their monotonicity. In this case, investors can long the best performer and short the worst performer so as to achieve excess returns. All indicators of the momentum-reversal strategy fail to maintain their monotonic patterns, indicating that this strategy is ineffective. The CMRS contains certain robustness as it achieves favorable results during a momentum crash.

⁸ The empirical results observed between Jan. 2015 and Aug. 2015 are opposite to those observed in 2018 although both periods are relatively stable. A potential reason is the existence of bubbles in 2015. The theory of rational bubbles (Blanchard and Watson, 1982) implies that investors' irrational behavior and expectations induce the stock price to deviate from its true value, giving rise to a bubble. Once the bubble bursts, it automatically leads to a crash in the market. Scheinkman and Xiong (2003) also point out that heterogeneous beliefs generated by investors' overconfidence bring bubbles to the market and then increase the possibility of its crash. Heterogeneous beliefs from overconfident investors would result in prices that deviate from the true value, and then lead to a momentum reversal effect in the market. As a result, the order of cumulative returns from quintiles 1 to 5 in Figure A.1 before the crash in August is different from that in Figure 1.

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